HHT BAO Locking — Results 2V0

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Sliding-τ Locking (IMF2)

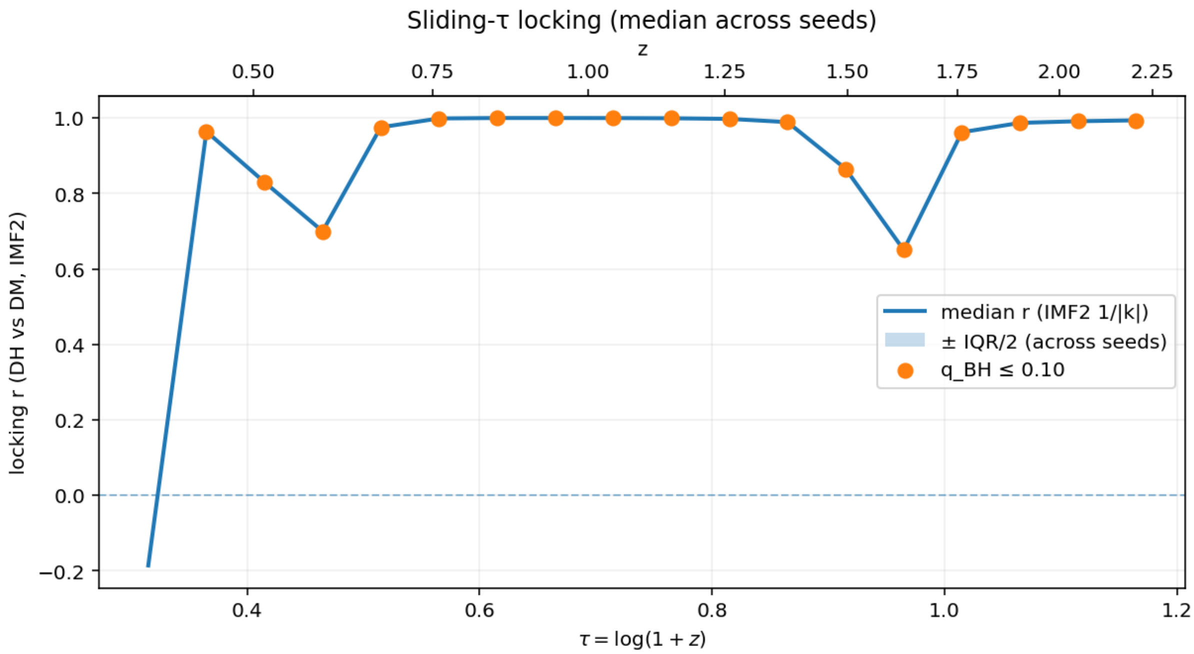

Figure: Median locking coefficient r between DH and DM 1/|k| (IMF2) in sliding τ-windows (width=0.25, step=0.05). Orange markers indicate windows passing BH-FDR q≤0.10.

Takeaways:
• r≈1 from τ≈0.55–1.15 (z≈0.75–2.2), indicating near-perfect wavelength locking over most of the range.
• Local dips appear near τ≈0.47 and τ≈0.96; still significant after FDR.
• Shaded band (not visible in this export) corresponds to ±IQR/2 across seeds; dispersion is small.

Locking r per Seed (Consistency Check)

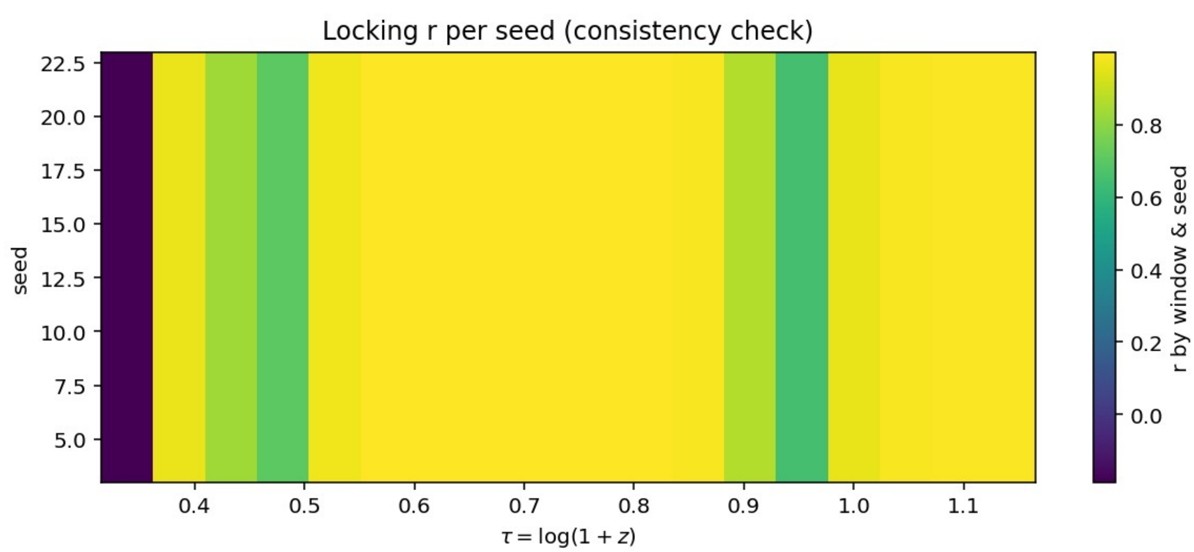

Figure: Color map of locking r by window (x-axis, τ) and seed (y-axis).

Takeaways:
• High r is consistent across all seeds in most windows (yellow band), supporting robustness.
• The same τ regions that dip in the sliding-τ curve also show slightly lower r here, matching the summary.

Distance-Gated Δr (Specificity Test)

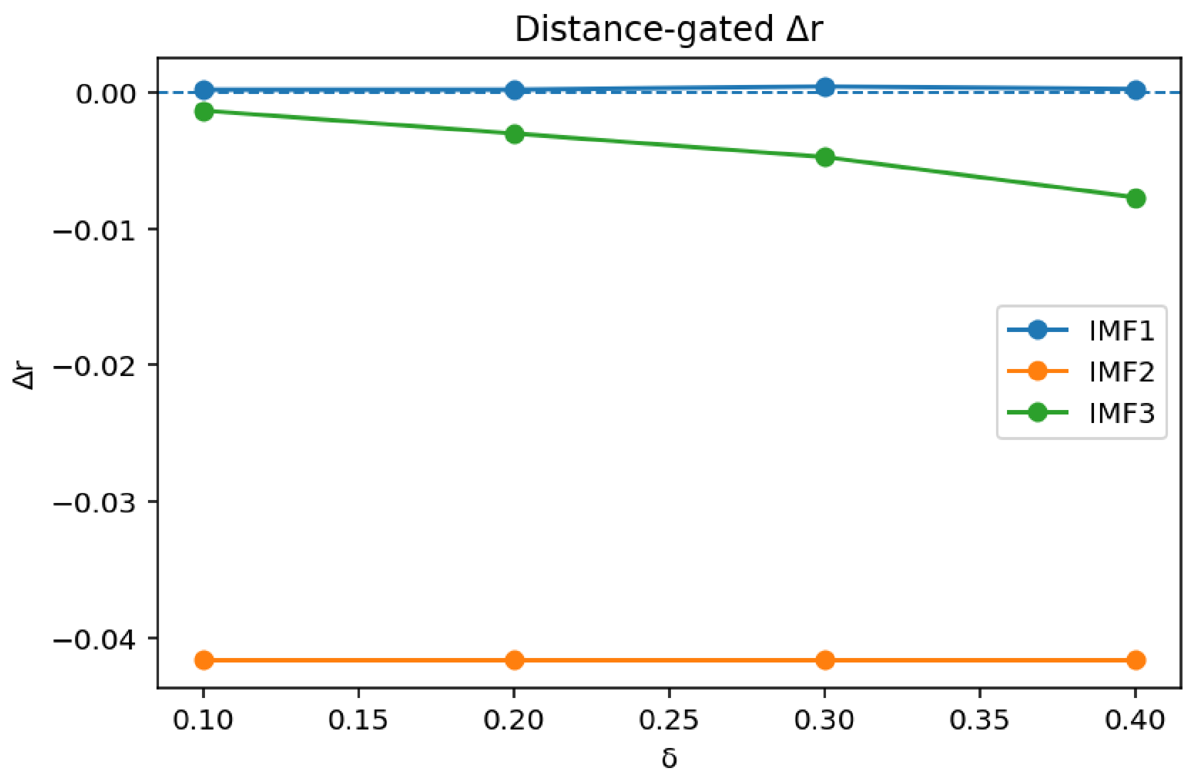

Figure: Change in r (Δr = gated − full) as points increasingly close to the IMF2 instantaneous frequency (k2) are excluded: δ ∈ {0.1, 0.2, 0.3, 0.4}.

Interpretation:
• IMF2: Δr≈−0.042 and nearly flat vs δ ⇒ removing near-k2 samples lowers r substantially and uniformly, consistent with locking being concentrated around the BAO-scale frequency.
• IMF1: Δr≈0 (insensitive to gating) ⇒ little evidence that IMF1 contributes to the locking.
• IMF3: small negative trend with δ (few×10^{-3}), indicating weak proximity-driven contribution compared to IMF2.

Method Invariance: Δr (IMF1)

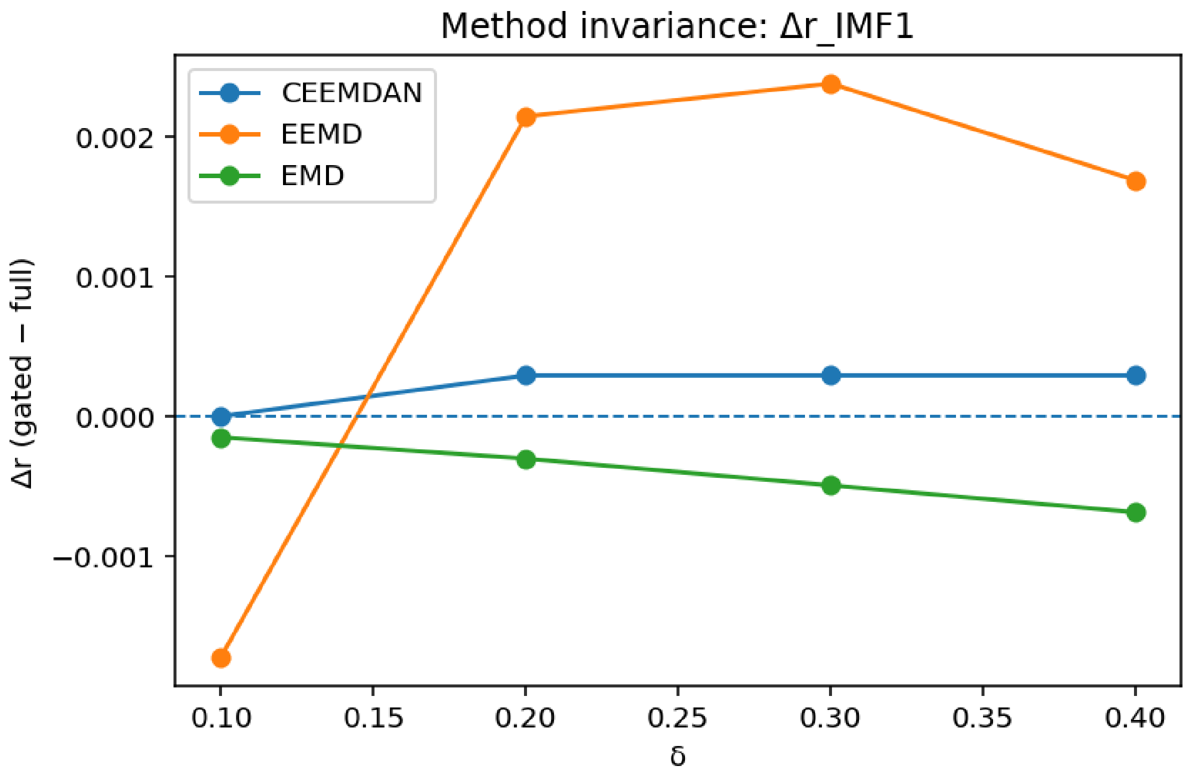

Figure: Distance-gated Δr for IMF1 across CEEMDAN, EEMD, and EMD at (pad, ngrid)=(0.15, 384).

Key points:
- All methods yield $|\Delta r| \lesssim 2\text{–}3\times10^{-3}$.
- Signs differ slightly (EEMD > 0, EMD < 0, CEEMDAN ≈ 0), but magnitudes are tiny ⇒ method choice does not materially change IMF1 specificity.

Method Invariance: Δr (IMF3)

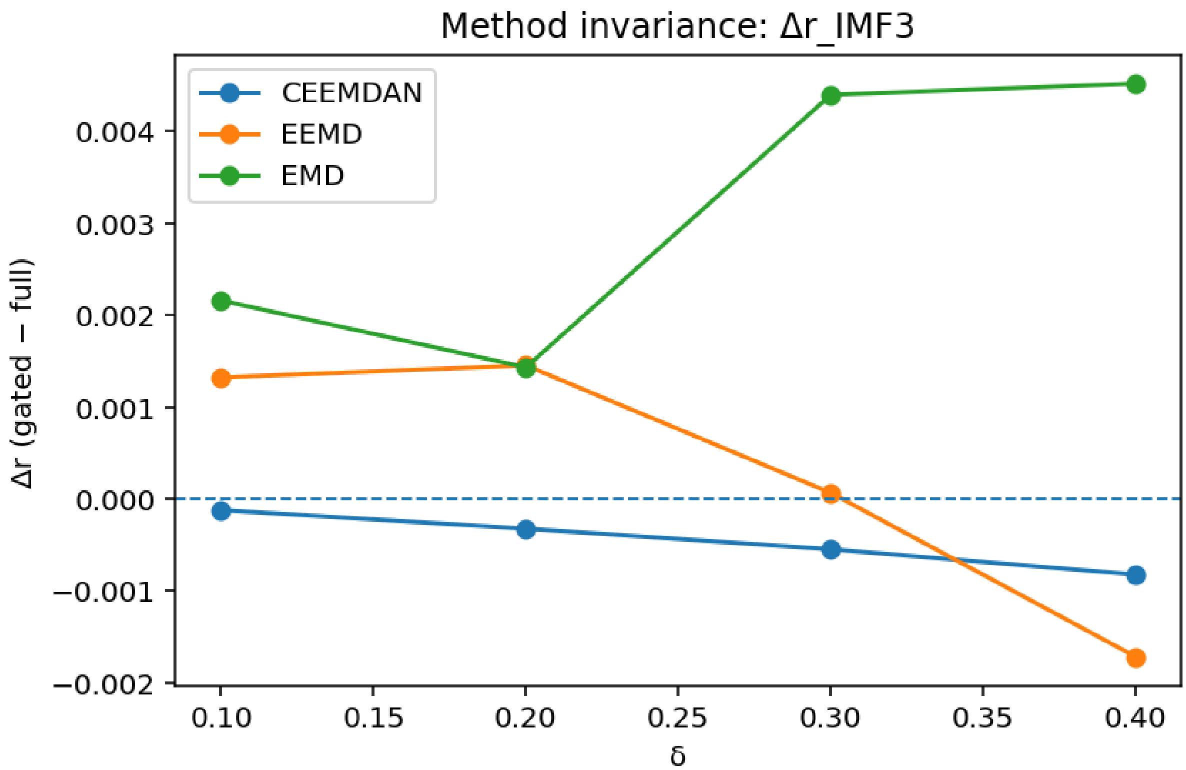

Figure: Distance-gated Δr for IMF3 across CEEMDAN, EEMD, and EMD at (pad, ngrid)=(0.15, 384).

Key points:
• CEEMDAN: small negative Δr ($\approx -10^{-3}$).
• EEMD: Δr transitions from $\approx +1.3\times10^{-3}$ to $\approx -1.7\times10^{-3}$ as δ increases.
• EMD: small positive Δr that grows with δ ($\approx +2\times10^{-3}$ to $+4.5\times10^{-3}$).
Overall magnitudes remain $O(10^{-3})$, much smaller than IMF2's effect.

Acronyms, Symbols, and Definitions

• HHT — Hilbert–Huang Transform; EMD/EEMD/CEEMDAN are its decomposition variants.
• IMF — Intrinsic Mode Function. IMF2 denotes the second mode; IMF1/IMF3 are neighboring modes.
• $k(\tau)$ — instantaneous angular frequency from the analytic signal phase; $1/|k|$ is the instantaneous wavelength proxy.
• DH / DM — data subsets (e.g., halves or disjoint selections) compared for locking.
• $\tau = \log(1+z)$ — uniformizing variable; sliding windows are defined in τ.
• r — Pearson correlation between DH and DM time series of $1/|k|$ within a window (locking coefficient).
• Δr — difference after applying a distance gate: ($r_{gated} - r_{full}$).
• Distance gate — keep samples with $|k_{imf} - k2|/|k2| \geq \delta$ (excludes points too close to the IMF2 frequency).
• BH-FDR (q) — Benjamini–Hochberg false discovery rate control applied across τ-windows.